

Task Management Application

Comprehensive Project Overview & Technical Documentation

Live App URL:	https://task-management-app-e305a.web.app
GitHub Repository:	https://github.com/prakashkmb12-afk/task_management_system

1. Project Executive Summary

The Task Management Application is a lightweight, responsive single-page web app built to enable authenticated users to seamlessly create, track, and manage personal task workflows in real time. Built with zero build-pipeline dependencies to guarantee 100% execution reliability.

2. Core Application Features

- **Google Authentication:** Secure single-click login powered by Firebase Auth (Google OAuth 2.0 popup).
- **Task Creation:** Enables creating tasks with a mandatory title and optional detailed description.
- **Real-Time Task View:** Automatically displays tasks scoped exclusively to the logged-in user, updated in real time via Cloud Firestore listeners.
- **Status Lifecycle Management:** Dynamic status tracking across three distinct states (Planned, In Progress, Complete).
- **Interactive Status Filtering:** Instant tab-based filtering (All, Planned, In Progress, Complete) for clear task organization.

3. Technical Stack & Architecture

Layer	Technology Selected	Rationale & Benefits
Frontend UI	HTML5 & CSS3 (Vanilla)	Zero build-step failures, fast load times, responsive cards.
Logic & SDK	JavaScript (ES Modules)	Modern modular imports directly via official Firebase CDN.
Authentication	Firebase Google Auth	Industry standard OAuth popup flow with session management.
Database	Cloud Firestore	NoSQL cloud database with real-time listener (onSnapshot).
Hosting	Firebase Hosting	Global CDN deployment with SSL encryption.

4. Task Workflow & Data Lifecycle

Tasks undergo a strict data lifecycle to ensure consistency, security, and immediate persistence.

Status State	Default / Trigger	Description & Behavior
Planned	Default State	Assigned automatically when a new task is created by the user.
In Progress	User Action	Selected when work has actively started on the task item.
Complete	User Action	Selected when work is finished. Visual badge turns green.

5. Data Isolation & Security Rules

To ensure each user can access **only their own tasks**, data isolation is enforced at both the client query layer and backend security rule layer:

- **User ID Binding:** Every created task document is stored with `userId: currentUser.uid`.
- **Client Query Scope:** Firestore queries fetch documents using `where('userId', '==', currentUser.uid)`.
- **Backend Rule Enforcement:** Firestore security rules restrict document access exclusively to authenticated users matching the document UID.

6. Support & Error Handling Matrix

Potential Condition	App Reaction & Recovery
User cancels Google popup	Displays clean banner: 'Sign-in popup was closed or interrupted.'
Double click sign-in	Button is automatically disabled during pending auth to prevent duplicate requests.
Empty task title	Form validation catches empty input and prevents database submission.
Firestore network error	Displays informative error banner without crashing the application UI.

7. Final Deployment Summary

The application is fully built, tested, and deployed to production on Firebase Hosting. All source code is version-controlled and public on GitHub.